

From spectra to univalence: a first tour of homotopy type theory

César Galindo

Expository note — written for a reader who already thinks in terms of spaces, homotopies, and fibrations (as one does after a thesis on orthogonal spectra), meeting type theory for the first time.

AIM. To build up the basic notions of Homotopy Type Theory — identity types, the univalence axiom, mere propositions, and higher inductive types — entirely as translations of things a homotopy theorist already knows: paths, homotopies of maps, contractibility, and a CW-type presentation of the circle. The payoff, exactly as in [UFP13, §8.1], is a synthetic computation of $\pi_1(S^1)$ that never mentions points, open sets, or a model of space at all.

1. TYPES AS SPACES

The guiding dictionary of [UFP13, §2.1] is: a *type* A behaves like a space, a *term* $a : A$ behaves like a point of that space, and a *dependent type* $B(x)$ for $x : A$ behaves like a fibration $B \rightarrow A$. Under this dictionary, equality itself becomes homotopical: for $x, y : A$ there is not a proposition but a *type* $x =_A y$, whose terms are witnesses that x and y agree — to be read as *paths* from x to y in the space A .

Definition 1.1 (Identity type, [UFP13, §1.12]). For a type A and $x, y : A$, the *identity type* $x =_A y$ is the type of paths from x to y . Every $x : A$ has a canonical path $\text{refl}_x : x =_A x$.

Theorem 1.2 (Path induction, [UFP13, §1.12]). *Let A be a type and let*

$$D : \prod_{x, y : A} (x =_A y) \rightarrow \mathcal{U}$$

be a family of types indexed by pairs of points of A together with a path between them. If there is a term $d(x) : D(x, x, \text{refl}_x)$ for every $x : A$, then there is a term $f(x, y, p) : D(x, y, p)$ for every $x, y : A$ and every $p : x =_A y$, with $f(x, x, \text{refl}_x) \equiv d(x)$.

Path induction says exactly that, to construct something along an arbitrary path, it suffices to construct it along the constant path. Every basic homotopy-theoretic fact about identity types — symmetry, transitivity, functoriality of paths under maps — is proved this way, mirroring how one proves facts about a space by proving them on a contractible neighborhood of the diagonal.

2. FUNCTION EXTENSIONALITY AND THE UNIVALENCE AXIOM

Two functions that agree pointwise are, in ordinary mathematics, simply equal; in type theory this has to be an axiom, and it turns out to be a special case of something much stronger.

Definition 2.1 (Equivalence, [UFP13, §4.1, §4.4]). A function $f : A \rightarrow B$ is an *equivalence* if its homotopy fibers $\text{fib}_f(b) := \sum_{a:A} (f(a) =_B b)$ are contractible for every $b : B$. Write $A \simeq B$ for the type of equivalences between A and B .

For every type A there is a canonical equivalence $\text{id}_A : A \simeq A$, hence a canonical map

$$\text{idtoeqv} : (A =_{\mathcal{U}} B) \rightarrow (A \simeq B)$$

sending refl_A to id_A , for any two types A, B in a universe $\mathcal{U}$.

Axiom 2.2 (Univalence, [UFP13, §2.10]). *For every universe $\mathcal{U}$ and all $A, B : \mathcal{U}$, the map $\text{idtoeqv} : (A =_{\mathcal{U}} B) \rightarrow (A \simeq B)$ is itself an equivalence.*

Informally: *equivalent types are equal*, and equal in a way that is itself faithfully recorded by the equivalence one started with. This is Voevodsky’s univalence axiom, the one that gives this repository its name and this document’s subtitle: it is the single extra principle that turns bare type theory into a foundation where isomorphism invariance is built in at the syntactic level, rather than imposed afterwards by fiat.

Theorem 2.3 ([UFP13, §2.9, §4.9]). *Univalence (Axiom 2.2) implies function extensionality: for $f, g : \prod_{x:A} B(x)$, the canonical map $(f = g) \rightarrow \prod_{x:A} (f(x) = g(x))$ is an equivalence.*

So the pointwise-equality principle a working mathematician reaches for instinctively is not a separate axiom bolted onto type theory: it falls out of univalence, the same way many closure properties of weak equivalences in a model category fall out of one saturation axiom rather than needing to be checked case by case.

3. MERE PROPOSITIONS

Not every type behaves like a genuine “space with interesting paths”; some behave like ordinary logical propositions, with at most one path between any two of their points.

Definition 3.1 ([UFP13, §3.3, §3.11]). A type A is a *mere proposition*, $\text{isProp}(A)$, when $\prod_{x,y:A} (x =_A y)$ is inhabited. It is *contractible*, $\text{isContr}(A)$, when $\sum_{a:A} \prod_{x:A} (a = x)$ is inhabited: a mere proposition that is, in addition, inhabited.

Remark 3.2. Definition 2.1 can now be read literally: f is an equivalence exactly when each of its fibers is a contractible type, i.e. a space that is homotopy equivalent to a point — precisely the fiberwise condition a homotopy theorist would write down for a weak equivalence of spaces or of orthogonal spectra.

4. A HIGHER INDUCTIVE TYPE: THE CIRCLE

Ordinary inductive types are generated by point constructors. A *higher* inductive type is also allowed path constructors — generators of its identity type — which is exactly how one builds genuinely homotopical spaces synthetically.

Definition 4.1 (The circle, [UFP13, §6.3–6.4]). The type S^1 is generated by

- a point **base** : S^1 , and
- a path **loop** : **base** $=_{S^1}$ **base**,

together with an induction principle: to construct a section of a family $D : S^1 \rightarrow \mathcal{U}$, it suffices to give a point $d_{\text{base}} : D(\text{base})$ together with a path $d_{\text{base}} =_{D(\text{base})} d_{\text{base}}$ lying over **loop**.

No quotient of an interval, no CW attaching map, no point-set model is needed to say this: the circle is presented purely by what one is allowed to induct on.

5. PAYOFF: $\pi_1(S^1) \simeq \mathbb{Z}$ SYNTHETICALLY

Theorem 5.1 ([UFP13, Thm. 8.1.1]). *$(\text{base} =_{S^1} \text{base}) \simeq \mathbb{Z}$, and hence $\pi_1(S^1, \text{base}) \cong \mathbb{Z}$ as groups.*

Remark 5.2 (Sketch of the encode–decode method, [UFP13, §8.1]). The proof constructs a family **code** : $S^1 \rightarrow \mathcal{U}$ by the recursion principle of Definition 4.1, sending **base** to $\mathbb{Z}$ and transporting along **loop** via the successor equivalence $\mathbb{Z} \simeq \mathbb{Z}$; univalence (Axiom 2.2) is exactly what makes this transport along a *loop* land back on $\mathbb{Z}$ rather than on some new, unnamed type. Maps **encode** : $(\text{base} = \text{base}) \rightarrow \text{code}(\text{base})$ and **decode** in the other direction are then shown to be mutually inverse, by path induction (Theorem 1.2) on one side and induction on $\mathbb{Z}$ on the other. Full details, including the fiberwise statement over all of S^1 that makes the induction go through, are in [UFP13, §8.1].

Remark 5.3 (Why this belongs next to a thesis on orthogonal spectra). Computing $\pi_1(S^1)$ is, of course, elementary in classical homotopy theory. What is new here is not the answer but the method: the computation above never leaves type theory to invoke a topological model of the circle, a simplicial set, or a spectrum — it is carried out entirely by induction principles internal to the type S^1 itself. The spectral-sequence techniques of `Theses_Galindo.tex` compute homotopy invariants *of* a fixed point-set or simplicial model; univalent foundations instead asks what invariants can be defined *without ever choosing a model at all*. Whether, or how, the orthogonal tower of that thesis admits a synthetic, univalent presentation is not addressed here — it is left as an open question, not a claim.

6. TAKEAWAYS

- Equality between points of a type is itself a type, built by the identity-type former, and everything about it is proved by path induction (Definition 1.1 and Theorem 1.2).
- Univalence says equivalent types are equal, and it alone forces function extensionality as a theorem rather than a separate postulate (Axiom 2.2 and Theorem 2.3).
- Mere propositions and contractible types (Definition 3.1) let Definition 2.1’s fiberwise definition of equivalence read exactly like a homotopy theorist’s definition of weak equivalence.
- Higher inductive types generate spaces by their induction principle alone; the circle (Definition 4.1) is the simplest nontrivial example.
- The synthetic computation $\pi_1(S^1) \simeq \mathbb{Z}$ (Theorem 5.1) is the standard first demonstration that this machinery computes real homotopy-theoretic invariants without a background model of space.

REFERENCES

- [UFP13] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. Book version `first-edition-697-gcffb6ea`. <http://homotopytypetheory.org/book/>.